Endpoint-Calibrated Graph-Native Tumor-State Forecasting with Probabilistic Functional Tumor Volume Readouts in Breast DCE-MRI

Iris Seaman, Tamer Oraby, and Murad Moqbel

***Abstract*— Serial dynamic contrast-enhanced magnetic resonance imaging during neoadjuvant breast cancer therapy provides repeated measurements of tumor burden, but most forecasting pipelines collapse this sequence to scalar functional tumor volume. This study presents endpoint calibration for graph-native probabilistic tumor-state forecasting, with functional tumor volume treated as a calibrated readout from the predicted graph rather than the only forecasted object. Each tumor is represented as a registration-consistent supervoxel graph, a scheduled-sampling message-passing forecaster predicts regional motion, imaging-feature evolution, and active status, and endpoint calibration aligns the graph rollout with future functional tumor volume. Empirical residual Monte Carlo simulation then converts the graph center into patient-level predictive distributions. Controlled synthetic graph tumors test when neighborhood information should help, showing that graph message passing is most useful when local neighborhoods reveal hidden spatial response structure. On 758 held-out patients from combined I-SPY2 and American College of Radiology Imaging Network 6698 (ACRIN-6698) graph-bearing breast magnetic resonance imaging resources, the final hybrid-edge model reduced deterministic functional tumor volume error from 12.88 mL to 4.63 mL, Monte Carlo mean error from 15.26 mL to 5.56 mL, and conformal 90% interval width from 59.51 mL to 22.38 mL, while improving raw 90% coverage from 0.766 to 0.877. The model also preserved sampled tumor-cloud geometry and identified low residual imaging burden at the final visit with area under the receiver operating characteristic curve 0.957. These results support structured longitudinal tumor-state forecasting with separate graph-dynamic, endpoint-calibration, and residual-uncertainty layers.**

***Index Terms*— Breast dynamic contrast-enhanced magnetic resonance imaging, conformal prediction, functional tumor volume, graph neural networks, longitudinal imaging, Monte Carlo simulation, uncertainty calibration.**

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I. Introduction

Serial dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) during neoadjuvant breast cancer therapy creates a longitudinal forecasting problem: given the tumor observed so far, estimate future active burden and the uncertainty around that burden. I-SPY established serial breast MRI as a response-monitoring substrate for adaptive neoadjuvant trials [1], and functional tumor volume (FTV) became a central quantitative MRI biomarker for response and recurrence risk [2]–[5]. Public I-SPY2 and American College of Radiology Imaging Network 6698 (ACRIN-6698) resources distributed through The Cancer Imaging Archive make the forecasting problem testable at cohort scale [6], [7].

Most predictive imaging pipelines reduce the serial tumor state to a scalar endpoint or a response label. That is clinically useful, but it discards the regional organization visible in DCE-MRI: local enhancement, washout, necrosis, boundary motion, and segmentation uncertainty can vary across the tumor. A graph of transported supervoxel regions gives each local region its own state while preserving spatial relationships. The forecasting question becomes how regional tumor states move, change enhancement, and remain active or become inactive over future visits.

This paper studies a layered alternative to scalar-only FTV regression. A scheduled-sampling graph forecaster rolls a registration-consistent tumor graph forward in time; endpoint calibration aligns the graph rollout with observed FTV and active tumor burden; empirical residual Monte Carlo (MC) then converts the calibrated center into patient-level FTV distributions. This separates three properties that are often conflated: spatial tumor-state plausibility, endpoint burden calibration, and predictive-distribution calibration.

The resulting contribution is a graph-native forecasting framework that keeps the tumor state structured while calibrating the FTV readout. Controlled synthetic tumors first establish when message passing should help: local neighborhoods must carry response information not fully visible from each node's own features. The held-out I-SPY2/ACRIN experiments then show that endpoint calibration and edge-aware neighborhoods improve FTV error, interval sharpness, coverage, and graph-state burden while preserving sampled tumor-cloud geometry. Strong scalar and hybrid FTV controls are retained to define the boundary of the claim, so the story is not that graph models replace scalar readouts universally; it is that graph structure is useful when the forecasted object includes the future tumor state and its calibrated burden.

II. Related Work and Positioning

Longitudinal breast MRI response models have used hand-crafted MRI features, radiomics, convolutional models, transformers, and temporal fusion networks to predict pathologic complete response or related response endpoints [4], [8], [9]. These studies motivate serial MRI representation learning, but they usually return a label or scalar endpoint rather than a future spatial tumor state. Digital twin work is the closest conceptual neighbor: image-calibrated mathematical models can simulate patient-specific breast tumor response under treatment [10]–[12]. The present study follows a complementary data-driven route focused on longitudinal MRI burden forecasting across observed visits.

These neighboring literatures motivate but do not replace the present task. Response classifiers are evaluated against pathology labels, mechanistic twins simulate patient-specific response with explicit tumor-growth models, and scalar FTV models provide strong endpoint readouts. The target here is the intersection that those approaches do not directly provide: a learned graph-native future tumor state with a calibrated patient-level FTV distribution.

Graph neural networks are natural for structured dynamical systems because message passing can aggregate local relational information [13], [14]. Scheduled sampling reduces exposure bias in autoregressive sequence prediction [15]. Here, these ideas are applied to transported tumor supervoxels: nodes represent local tumor regions, edges represent local spatial or feature-neighborhood relationships, and the model predicts the next graph state.

The uncertainty layer is empirical rather than Bayesian. Model weights are fixed after training; uncertainty samples are generated by resampling held-out residual structure around the deterministic rollout center. The resulting FTV distributions are evaluated with mean error, coverage, conformal intervals [16], [17], and Continuous Ranked Probability Score (CRPS), a proper score for predictive distributions [18]. This makes the paper a probabilistic imaging-burden forecasting study rather than a response-classification or segmentation study.

III. Methods and Evaluation Design

The method is organized around three separable layers. The graph rollout predicts a future tumor state; endpoint calibration aligns the graph readout with aggregate FTV and active burden; residual MC and conformal calibration convert the fixed rollout center into a patient-level predictive distribution. The evaluation keeps these layers separate so scalar FTV accuracy does not substitute for graph-state validity.

A. Forecasting Task and Graph Representation

For patient p and visit t, the tumor graph is $G_p^{(t)} = (V_p^{(t)}, E_p^{(t)})$. Each node is a compact baseline tumor supervoxel produced by Simple Linear Iterative Clustering (SLIC) [19] and transported to follow-up visits by symmetric diffeomorphic registration [20]. Node features are $\log(1 + \text{voxel count})$, volume in mL, percent enhancement (PE) mean and standard deviation, signal enhancement ratio (SER) mean and standard deviation, and centroid position after subtracting the visit-level tumor centroid. The active target records whether the transported supervoxel has tumor support at that visit.

The fixed baseline partition is used to preserve node identity across visits: the same baseline region can shrink, move under registration, change enhancement, or become inactive at later visits. This representation separates local state evolution from the scalar burden readout. A patient with complete T_0–T_3 graph support therefore contributes a sequence of aligned node states rather than four unrelated segmentations.

The forecasted object is a future graph state $\widehat{G}_{p,t}$ conditioned on observed visits $G_{p,\le s}$. FTV is read from that state as

$$\widehat{\mathrm{FTV}}_{p,t} = \sum_{i \in V_p^{(t)}} \widehat{a}_{p,t,i} \widehat{v}_{p,t,i}, \tag{1}$$

where $\widehat{a}_{p,t,i}$ is the predicted active probability and $\widehat{v}_{p,t,i}$ is the predicted node volume. Spatial quality is therefore evaluated separately from scalar burden by sliced Wasserstein distance (SWD), Chamfer distance, Dice, and active-node-count error.

B. Graph Rollout and Endpoint Calibration

All graph-family models use the same autoregressive local spatial graph convolution backbone. The forecaster embeds node features into a 64-dimensional hidden state, applies residual message passing with shared weights, and predicts next-visit displacement, feature change, and active-node logit. During training, scheduled sampling transitions from teacher-forced history to self-fed rollouts. The original scheduled-sampling graph rollout is the spatial-state baseline before endpoint calibration is added.

The rollout predicts changes rather than treating each future visit as an independent endpoint. Predicted displacement updates node centroids, feature-change heads update local imaging features, and active logits determine the probability that each transported supervoxel remains in the future tumor support. For multi-step forecasts, the predicted graph state is fed forward so later predictions depend on earlier model outputs.

Endpoint calibration adds losses on aggregate FTV and active-node burden while retaining the node-level spatial and feature losses. The first calibrated model uses spatial $k = 8$ neighborhoods and is denoted Endpoint+Active. The final retained model, Hybrid-Edge $k = 8$, rebuilds neighborhoods dynamically with a hybrid spatial-feature distance and radial imaging-feature edge attributes. Candidate neighbors are ranked by an equally weighted normalized blend of spatial and feature distance, then symmetrized. Edge attributes encode relative displacement and local enhancement/volume differences. The retained real-cohort graph uses a 64-dimensional hidden state, two message-passing layers, $k = 8$ neighborhoods, scheduled sampling annealed to 0.5, and endpoint weights $\lambda_{\mathrm{FTV}} = 0.20$ and $\lambda_{\mathrm{act}} = 0.05$.

Table I collects the retained settings and leakage controls that must stay fixed when the endpoint and uncertainty layers are evaluated.

TABLE I
RETAINED HYBRID-EDGE $k = 8$ CONFIGURATION AND LEAKAGE CONTROLS.

Component	Setting
Node identity	Baseline SLIC supervoxels transported to follow-up visits by registration.
Node state	Local volume, PE/SER summaries, centered centroid, and active probability.
Backbone	64 hidden channels, two residual message-passing layers, autoregressive rollout.
Edges	Symmetric hybrid spatial-feature $k = 8$ neighborhoods with radial imaging-feature attributes.
Training	Five held-out-patient folds; scheduled sampling annealed to 0.5.
Endpoint loss	$\lambda_{\text{FTV}} = 0.20$, $\lambda_{\text{act}} = 0.05$, active-logit loss retained.
MC calibration	128 residual draws per endpoint; target-patient residuals excluded.

For a rollout ending at visit t, the training objective combines local state losses with endpoint losses,

$$\mathcal{L} = \mathcal{L}_{\text{node}} + \lambda_{\text{FTV}} \left| \widehat{\text{FTV}}_{p,t} - \text{FTV}_{p,t} \right| + \lambda_{\text{act}} \left| \sum_i \widehat{a}_{p,t,i} - \sum_i a_{p,t,i} \right|. \tag{2}$$

The node term includes feature, displacement, and active-logit losses; the FTV term centers the scalar endpoint read from the graph; and the active-burden term prevents scalar FTV calibration from drifting away from the sampled active state. This design keeps the graph rollout interpretable: the scalar endpoint is not trained as an unrelated head, but as a readout constrained by predicted node volume and activity.

Model selection follows the same held-out-fold structure used for final evaluation. Candidate edge families are compared as matched endpoint-calibrated models so that the retained topology is not selected merely because it changes the loss family. The retained HYBRID-EDGE $k = 8$ is chosen as the graph-family model with the best balance of endpoint error, interval behavior, and sampled-cloud geometry rather than the single lowest scalar FTV error.

C. Residual MC and Evaluation Hygiene

After training, the deterministic graph rollout is fixed. Patient-level MC samples are generated by resampling held-out residuals from matching conditioning buckets, excluding the target patient's own residuals. The same MC protocol is used for graph, scalar, and hybrid centers. Scalar controls include last-observed FTV and regularized temporal FTV centers; the hybrid center fits an out-of-fold ridge model using the graph prediction, last-observed FTV, and T_0 FTV.

Each endpoint MC distribution uses 128 residual draws. For graph-family models, the deterministic rollout also provides a predicted active-probability vector and node volumes; residual draws perturb the endpoint FTV center and sampled active burden while preserving the patient-specific graph support. Graph-state metrics are therefore computed from predicted and observed active supervoxel clouds, whereas scalar controls are evaluated only on FTV distributions. This keeps the scalar readout comparison fair without pretending that scalar baselines produce a future tumor graph.

Residual buckets are indexed by conditioning visit and predicted visit. For each draw, an FTV residual is sampled from other held-out patients in the same bucket. For graph-family MC, active-count residuals perturb the predicted active mass, and active masks are sampled from the predicted node probabilities before cloud-level metrics are computed. Raw 90% intervals are the empirical 5th and 95th percentiles of the endpoint samples. Conformal intervals expand these raw intervals by a bucket-specific absolute nonconformity correction computed from other held-out patients.

All reported real-cohort estimates use held-out-patient folds. Residual pools, conformal calibration, scalar baselines, and hybrid readouts are fit without using the target patient's held-out residuals. This is important because the MC distribution can otherwise look calibrated by reusing a patient's own forecast error. The reported conformal widths are therefore computed after the same target-patient exclusion rule used for residual sampling.

Split files, preprocessing summaries, retained configurations, evaluation scripts, and figure/table generation code are maintained in the reviewer-facing release repository. Protected patient-level imaging data are not redistributed.

D. Baselines and Claim Boundaries

The ablation design follows the layered claim. It begins with the original scheduled-sampling graph rollout, which establishes the spatial-state forecast before endpoint losses are added. Comparing that rollout with ENDPOINT+ACTIVE tests whether aggregate FTV and active-burden losses recenter the graph readout without discarding the local tumor state. The next comparisons keep endpoint calibration fixed and vary the neighborhood design: no-edge, spatial, radial, feature-only, radial imaging-feature, and hybrid spatial-feature variants ask whether local relations add information beyond each node's own history. The final boundary is scalar-only forecasting. Last-observed, temporal ridge, kernel temporal, and hybrid graph-plus-scalar centers use the same residual-MC family to show how far scalar readouts can go when the requested output is only final-visit FTV.

These comparators are not expected to favor the graph model in every column. A scalar residual model can be sufficient or better for a scalar endpoint. The retained graph model is therefore judged on the joint requirement of producing a calibrated FTV distribution while preserving a future tumor graph whose active-node burden and sampled-cloud geometry can be evaluated.

Figure 1 summarizes this ordering from image-derived graph construction through rollout, endpoint readout, residual uncertainty, and separate endpoint and graph-state evaluation.

E. Cohorts, Synthetic Controls, and Metrics

The real-cohort evaluation uses 758 graph-bearing patients with four longitudinal DCE-MRI visits across the I-SPY2/ACRIN-derived cohort. Each patient is evaluated in the fold in which that patient was held out. The primary real-cohort task is $T_0 \rightarrow T_3$ FTV forecasting; the same final-visit target is also evaluated from T_1 and T_2 conditioning. Primary metrics are deterministic FTV mean absolute error (MAE), MC-mean FTV MAE, CRPS, raw 90% coverage, and interval width. Secondary

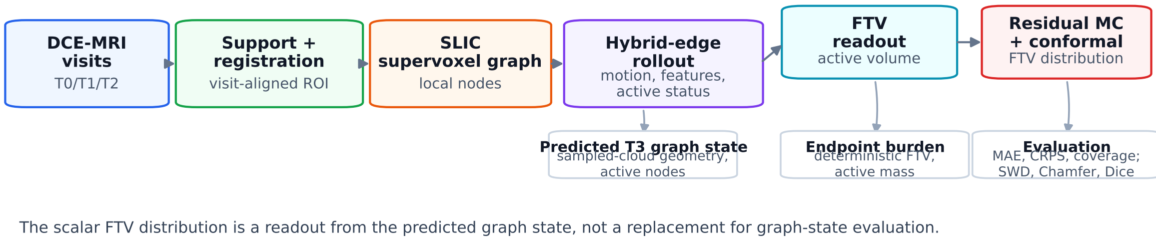

Fig. 1. End-to-end endpoint-calibrated spatial graph forecasting pipeline. Serial DCE-MRI visits are converted into a registration-consistent SLIC supervoxel graph, a hybrid-edge rollout predicts future node motion, imaging-feature evolution, and active status, FTV is read from predicted active volume, and residual MC with conformal correction produces the patient-level predictive distribution. Evaluation keeps endpoint metrics and graph-state metrics separate.

metrics include active-node-count error, SWD, Chamfer distance, Dice, source and subtype stratification, a small external Breast-MRI neoadjuvant chemotherapy pilot stress test, and low/high residual MRI-burden threshold readouts.

The primary endpoint is final-visit FTV because it is the scalar burden most commonly used for treatment-response monitoring in this setting. The graph-state metrics are reported alongside FTV because a graph rollout can be well centered in scalar burden while producing implausible spatial support, or spatially plausible while needing endpoint recentering. The evaluation therefore treats endpoint accuracy, distribution calibration, and graph-state plausibility as distinct claims.

For a patient-level predictive distribution $\widehat{F}$ and observed FTV y, CRPS is used as the proper scoring rule,

$$\mathrm{CRPS}(\widehat{F}, y) = \int_{-\infty}^{\infty} \left(\widehat{F}(z) - \mathbb{1}\{y \leq z\}\right)^2 dz. \tag{3}$$

Raw 90% coverage reports whether y falls between the empirical 5th and 95th percentiles of the MC samples. Conformal 90% coverage uses the expanded interval described above. These distributional scores are interpreted together: coverage alone can be improved by making intervals too wide, while CRPS and width penalize poorly centered or diffuse distributions.

Graph-state metrics are computed on active supervoxel clouds at the predicted visit. Active-count error measures whether the model predicts the correct amount of active tumor support. SWD compares projected point-cloud distributions and is less dominated by a few extreme nodes than a maximum-distance criterion. Chamfer distance measures nearest-neighbor geometric mismatch between predicted and observed active clouds. Dice is computed after voxelizing the predicted and observed active supports, giving a familiar overlap score for the forecasted tumor state. These metrics are intentionally reported even when scalar FTV is strong, because they test the part of the output that scalar baselines cannot produce.

Synthetic graph tumors are generated to test the graph assumption under known dependence structure before interpreting real-cohort ablations. The node-local regime makes future response mostly recoverable from each node's own features. The neighbor-coupled regime makes future response depend partly on local neighbor state. The latent spatial-field regime drives response from a hidden smooth regional field observed through noisy local proxies. The synthetic question is whether message passing improves endpoint and graph-state metrics when local neighborhoods reveal information that node-local features only partially observe.

All synthetic regimes use the same training, endpoint-calibration, and residual MC evaluation family as the real-cohort graph models. This keeps the synthetic study interpretive: a gain in the latent-field regime supports the premise that local neighborhoods can denoise hidden regional response structure, while a weak gain in the node-local regime sets the expectation that edges should help conditionally rather than universally. The real-cohort edge ablations are interpreted through this same lens.

The small external Breast-MRI neoadjuvant chemotherapy pilot (Breast-MRI-NACT-Pilot) analysis is treated only as a stress test of model-family transfer. It uses source-trained graph checkpoints and source-cohort residual MC calibration on 11 graph-ready external patients. No target-cohort residuals are used for recalibration, so the result tests relative ordering under collection shift rather than powered clinical generalization.

IV. Results

A. Synthetic Controls Identified When Edges Help

The synthetic results support a conditional view of message passing. When response was node-local, the no-edge model remained competitive for scalar FTV. When response depended on a latent spatial field, Hybrid-Edge $k = 8$ reduced $T_0 \to T_3$ MC-mean FTV MAE by 5.43 mL relative to the no-edge endpoint model, reduced CRPS by 3.53, narrowed raw 90% interval width by 24.14 mL, and improved MC Chamfer distance by 0.41 mm (Table II). The real-cohort experiments therefore test both endpoint calibration and whether edge design makes neighborhood information useful in breast MRI graphs.

Figure 2 visualizes the latent-field setting in which neighborhood information provides the clearest graph signal.

B. Endpoint Calibration and Edge-Aware Graphs Improved the Rollout

The scheduled-sampling graph rollout established a coherent spatial-state forecast, and endpoint calibration moved its FTV readout toward the observed final burden. In the held-out $T_0 \to T_3$ evaluation, observed mean FTV was 24.94 mL and the

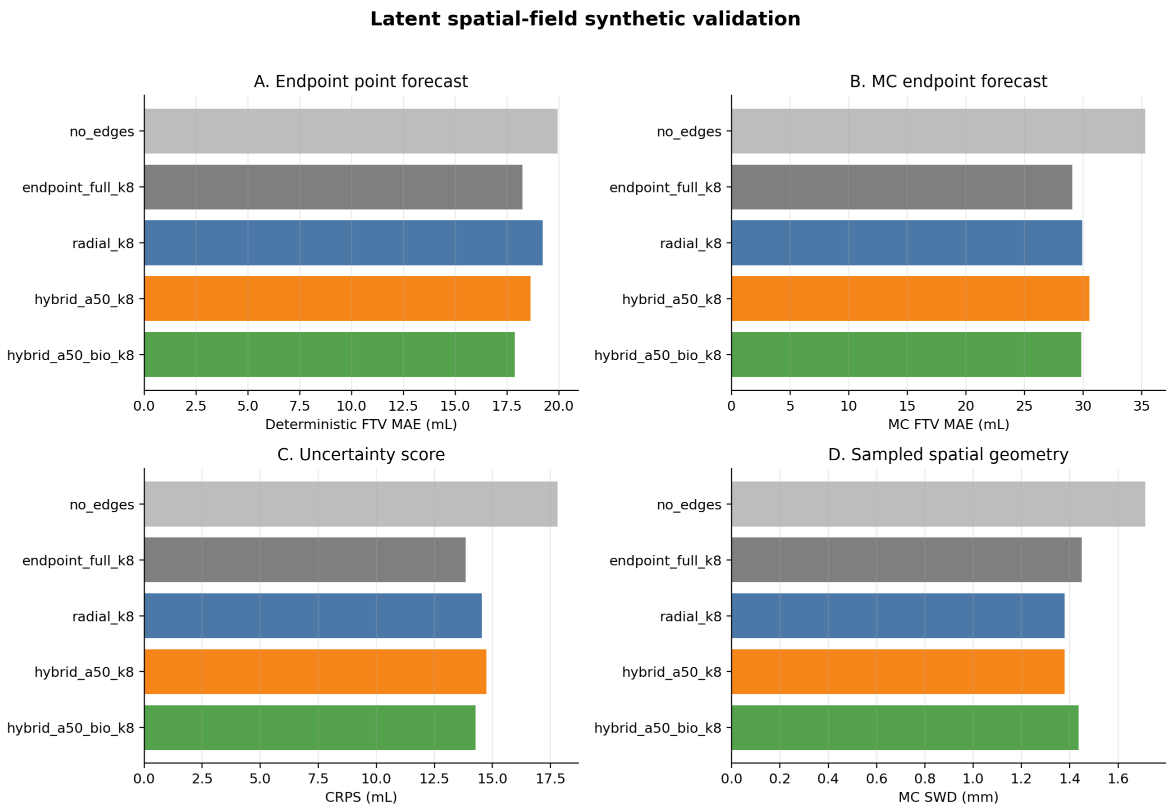

Fig. 2. Synthetic latent spatial-field validation. The retained graph becomes most useful when hidden regional response structure can be denoised through local neighborhoods, improving endpoint uncertainty and sampled-cloud geometry relative to the no-edge control.

TABLE II
SYNTHETIC $T_0 \rightarrow T_3$ MECHANISM CHECKS. DELTAS ARE HYBRID-EDGE $k = 8$ MINUS THE NO-EDGE ENDPOINT MODEL; NEGATIVE VALUES ARE BETTER FOR MAE, CRPS, WIDTH, AND CHAMFER.

Regime	Graph signal	Δ MC MAE (mL)	Δ CRPS	Δ width (mL)	Δ Chamfer (mm)
Node-local	Future response readable from node features.	+0.15	-0.04	+0.99	-0.47
Neighbor-coupled	Future response partly depends on local neighbors.	+0.40	-0.20	-1.02	-0.62
Latent field	Hidden smooth response field is observed through noisy node proxies.	-5.43	-3.53	-24.14	-0.41

baseline deterministic mean was 12.43 mL, yielding 12.88 mL MAE and -12.51 mL bias. Adding endpoint and active-burden calibration reduced deterministic MAE to 5.88 mL, bias to -1.73 mL, and active-node-count absolute error from 58.00 to 1.70 supervoxels, while SWD, Chamfer, and Dice changed only minimally. This shows that the spatial rollout and endpoint readout are separable.

Using the same residual MC sampler, endpoint calibration reduced MC-mean FTV MAE from 15.26 to 7.61 mL, increased raw 90% coverage from 0.766 to 0.876, reduced raw interval width from 56.87 to 24.42 mL, reduced conformal 90% interval width from 59.51 to 27.95 mL, and reduced CRPS from 8.35 to 5.16. The paired MC-mean MAE reduction was 7.65 mL (95% CI: 7.11–8.20).

The final edge-aware graph further improved the graph-family tradeoff. Compared with the no-edge endpoint model, HYBRID-EDGE $k = 8$ lowered deterministic FTV MAE by 0.65 mL, MC-mean FTV MAE by 1.02 mL, CRPS by 0.56, raw interval width by 3.67 mL, conformal width by 4.12 mL, and MC Chamfer by 0.043 mm. The radial imaging-feature model had the lowest scalar graph MC-mean MAE by a small margin, but HYBRID-EDGE $k = 8$ was retained because it gave the strongest balance of coverage, conformal width, and sampled-cloud geometry.

Table III reports the main held-out comparison and keeps scalar-only rows in the same display so the graph-state claim is read against strong endpoint controls.

Figure 3 shows two held-out graph-state diagnostics with nontrivial final burden, one from each source collection. These examples are not used as cohort-level performance evidence. They show the observed baseline and final tumor graphs, exact retained-model graph-state metrics at T_3, and the endpoint MC distribution for the same forecast. This makes the graph-native object visible while Tables III and IV carry the aggregate evidence.

The scalar controls define the readout boundary of the contribution. A last-observed FTV center with out-of-fold residual MC achieved 4.82 mL MC-mean MAE, 0.898 raw coverage, and 3.76 CRPS. The hybrid graph-plus-scalar center achieved 4.50 mL MC-mean MAE and 3.71 CRPS. Thus, scalar

TABLE III

$T_0 \rightarrow T_3$ MAIN COMPARISON. DET. IS DETERMINISTIC, WIDTH IS RAW 90% INTERVAL WIDTH FOR ROWS WITH MC SAMPLES, AND SV DENOTES SUPERVOXELS. ACTIVE ERROR IS MC ACTIVE-NODE-COUNT ABSOLUTE ERROR; SCALAR ROWS DO NOT PRODUCE GRAPH-NATIVE STATE OUTPUTS.

Model	Det. MAE (mL)	MC MAE (mL)	CRPS	Raw cov.	Raw width (mL)	Active err. (SV)	MC SWD (mm)	MC Chamfer (mm)
Original graph rollout	12.88	15.26	8.35	0.766	56.87	40.78	1.320	3.656
ENDPOINT+ACTIVE graph	5.88	7.61	5.16	0.876	24.42	2.69	1.207	3.485
No-edge endpoint graph	5.28	6.59	4.56	0.871	22.87	2.69	1.209	3.491
Radial MRI-feature graph	4.62	5.52	3.95	0.872	20.05	2.66	1.177	3.457
HYBRID-EDGE $k = 8$ graph	**4.63**	**5.56**	**4.00**	**0.877**	**19.21**	**2.74**	**1.172**	**3.449**
Last-observed scalar MC	4.81	4.82	3.76	0.898	18.60	–	–	–
Hybrid graph+scalar MC	4.48	4.50	3.71	0.897	18.20	–	–	–

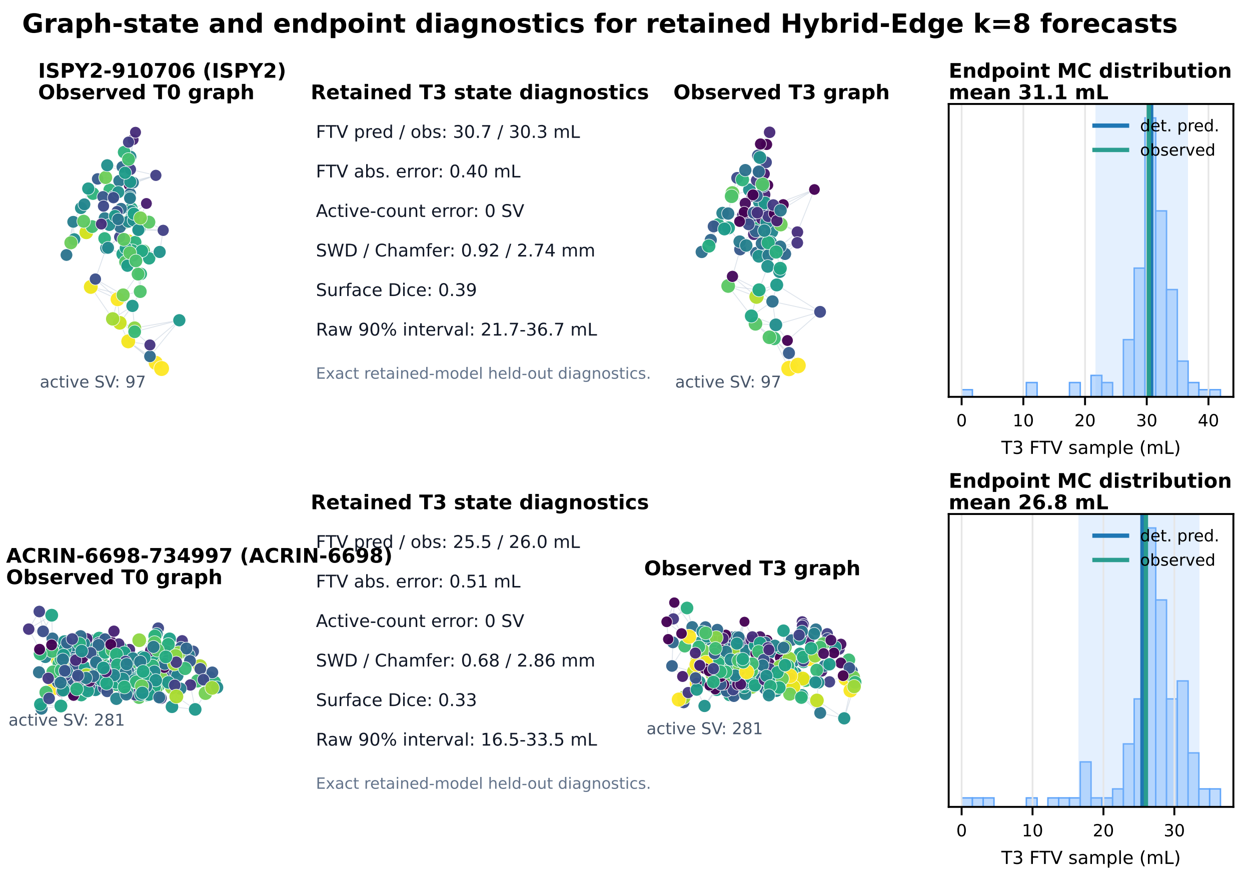

Fig. 3. Held-out graph-state and endpoint diagnostics from the retained HYBRID-EDGE $k = 8$ graph. Each row shows observed T_0 and T_3 active supervoxel graphs, exact retained-model T_3 graph-state diagnostics (FTV error, active-count error, SWD, Chamfer, and surface Dice), and the MC final-visit FTV distribution.

or hybrid calibration is strongest when the requested output is only final FTV. The graph rollout supplies a different object: a structured tumor-state forecast with active-node dynamics, sampled-cloud geometry, and a calibrated endpoint readout.

Figure 4 places the scalar, graph, and hybrid residual-MC distributions side by side for this readout-boundary comparison.

Figure 5 shows the corresponding reliability and probability integral transform diagnostics for the retained graph distribution.

Figure 6 shows the same improvement at the patient-distribution level by sorting endpoint intervals by observed final burden.

C. Robustness, Transfer Stress Test, and Clinical Readouts

The remaining analyses test whether this calibrated distribution remains stable across conditioning horizons, source collections, burden strata, and clinically interpretable MRI-burden thresholds.

The retained graph improvement persisted across conditioning horizons. MC-mean FTV MAE was 5.56, 5.90, and 5.68 mL for T_0, T_1, and T_2 conditioning, with paired reductions versus the original graph rollout of 9.70, 8.97, and 8.13 mL. Source-stratified internal robustness was stable: ACRIN-6698 had 4.55 mL MC-mean MAE and 0.881 raw coverage, while I-SPY2 had 5.93 mL MC-mean MAE and 0.876 raw coverage. In the small independent Breast-MRI-NACT-Pilot stress test,

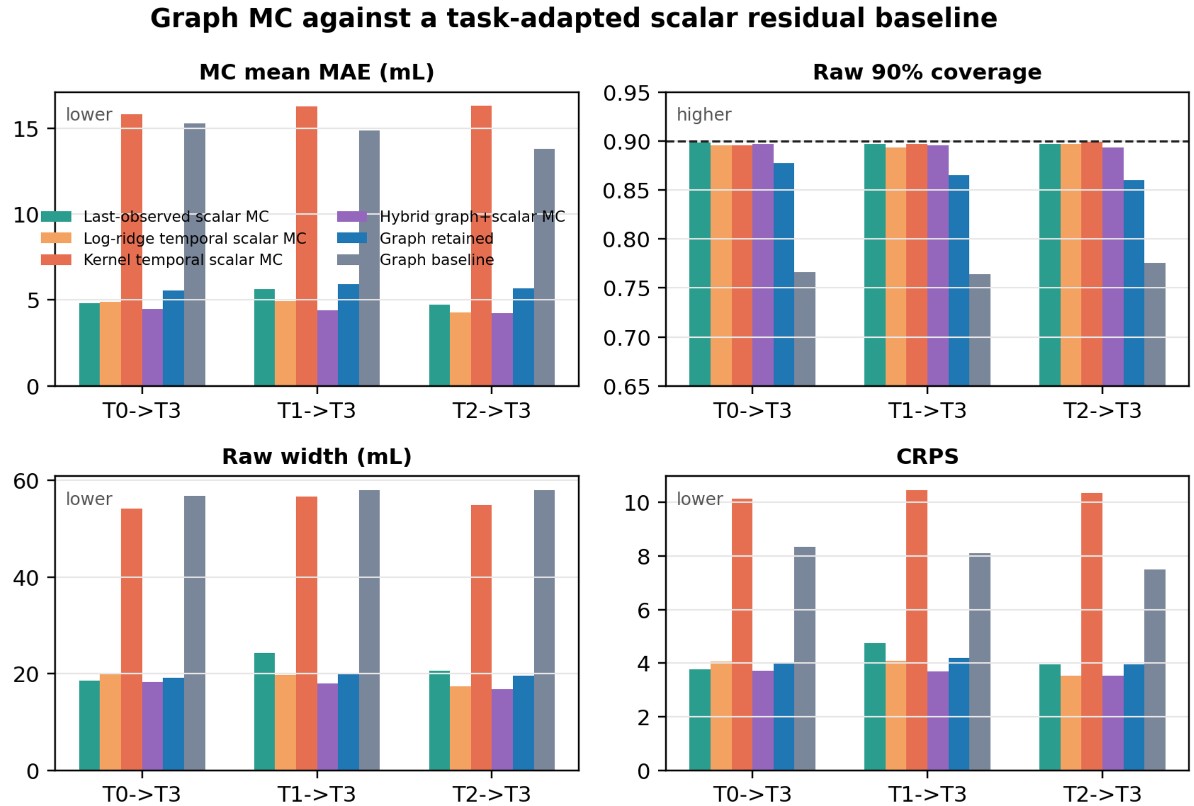

Fig. 4. Scalar, graph, and hybrid residual MC controls for final-visit FTV. Scalar temporal baselines use observed FTV history and metadata before applying out-of-fold residual MC. The hybrid model combines the graph center with scalar FTV history, showing that scalar readout calibration and graph-native state forecasting answer related but distinct questions.

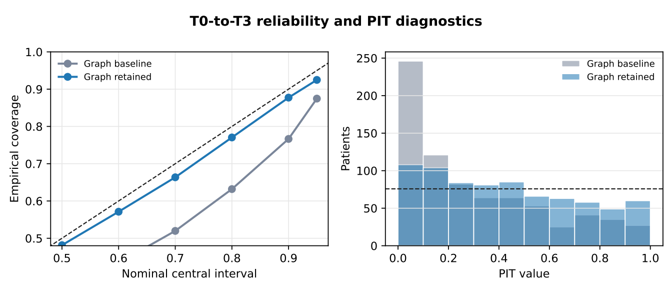

Fig. 5. $T_0 \to T_3$ reliability and probability integral transform (PIT) diagnostics. The retained graph improves central-interval coverage across nominal levels and reduces boundary concentration in the PIT distribution relative to the original graph rollout.

TABLE IV

COMPACT ROBUSTNESS AND BOUNDARY SUMMARY FOR THE RETAINED HYBRID-EDGE $k = 8$ GRAPH MC ROLLOUT. RAW COV. IS RAW COVERAGE AND CONF. WIDTH IS CONFORMAL 90% INTERVAL WIDTH. THE NEOADJUVANT CHEMOTHERAPY PILOT ROW IS A TRANSFER STRESS TEST WITH 11 GRAPH-READY PATIENTS, NOT A POWERED EXTERNAL VALIDATION COHORT.

Check	n	MC MAE (mL)	CRPS	Raw cov.	Conf. width (mL)
$T_0 \to T_3$ horizon	758	5.56	4.00	0.877	22.38
$T_1 \to T_3$ horizon	758	5.90	4.18	0.865	25.61
$T_2 \to T_3$ horizon	758	5.68	3.94	0.860	24.45
ACRIN-6698 source	202	4.55	2.93	0.881	21.78
I-SPY2 source	556	5.93	4.39	0.876	22.59
NACT external stress	11	15.01	–	0.545	–
Top 5% T_3 burden	38	31.57	–	0.237	–

HYBRID-EDGE $k = 8$ was best among the tested graph-family variants but still overpredicted FTV, giving 15.01 mL MC-mean MAE and 0.545 raw 90% coverage. The value of this analysis is conservative: without target-cohort residual recalibration, it tests whether model-family ordering persists under collection shift and identifies recalibration as the next validation step. Table IV summarizes these horizon, source, transfer, and tail checks in the same metric family used for the primary comparison.

Burden-tail behavior is the main safety boundary exposed by the uncertainty analysis. Below the top decile of observed T_3 FTV, HYBRID-EDGE $k = 8$ reached 3.97 mL MC-mean MAE, 0.924 raw coverage, and 2.49 CRPS; in the top 5% it reached 31.57 mL MC-mean MAE and 0.237 raw coverage. Molecular subtype checks showed improvement over the original graph rollout in every stratum, but the hormone receptor-negative/human epidermal growth factor receptor 2-positive (HR-/HER2+) group was smallest and hardest ($n = 57$, 7.52 mL MC-mean MAE).

Figure 7 shows this burden-conditional calibration directly, separating the aggregate improvement from the high-volume tail that needs targeted recalibration.

The MC samples also support clinically interpretable MRI-burden readouts. For low residual imaging burden, $P(T_3\,\text{FTV} < 5\ \text{mL})$ gave area under the receiver operating characteristic curve (AUC) 0.957, sensitivity 0.755, specificity

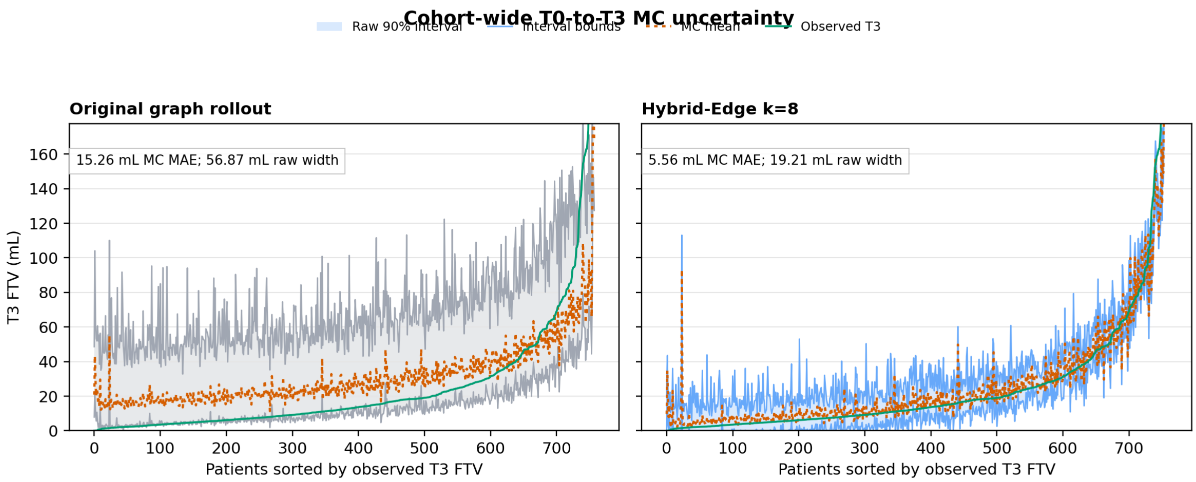

Fig. 6. Cohort-wide $T_0 \to T_3$ MC uncertainty bands sorted by observed final-visit FTV. The retained graph shifts MC means toward observed burden and narrows the patient-level predictive bands relative to the original graph rollout.

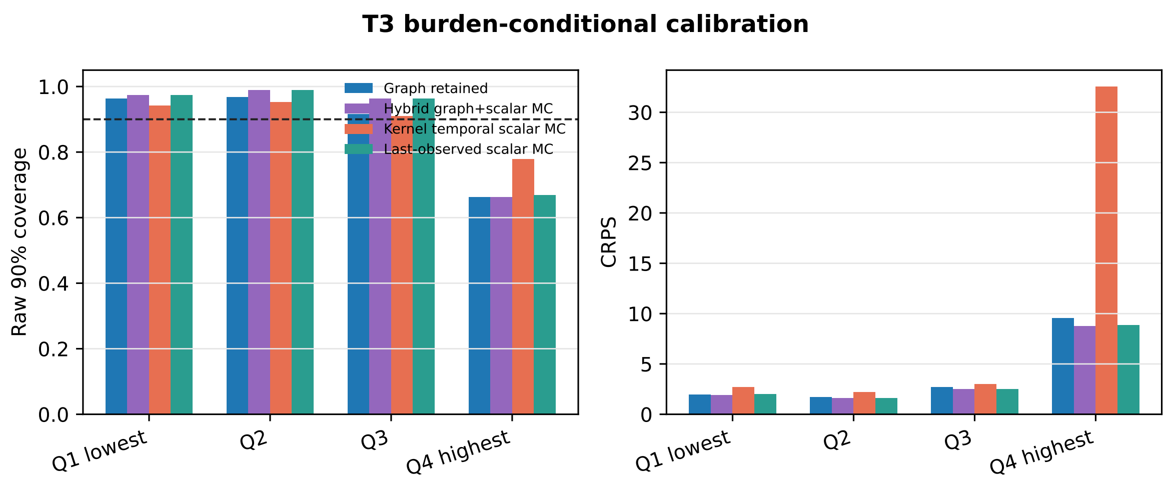

Fig. 7. Burden-conditional $T_0 \to T_3$ calibration by observed final-visit FTV quartile. The retained graph distribution improves over the original graph baseline but remains less calibrated in the high-volume tail; scalar and hybrid readouts remain strong when only scalar burden is required.

0.970, positive predictive value (PPV) 0.870, and negative predictive value (NPV) 0.937. For substantial residual burden, $P(T_3\,\mathrm{FTV} > 20\text{ mL})$ gave AUC 0.984, sensitivity 0.967, specificity 0.914, PPV 0.844, and NPV 0.983. Last-observed FTV was similarly discriminative, reinforcing that the added value of the graph framework is calibrated future burden plus graph-native tumor-state forecasting. Table V reports these threshold readouts, and Figure 8 shows how the same monitoring view changes as additional MRI visits are available for conditioning.

TABLE V

SECONDARY MRI-BURDEN THRESHOLD READOUTS FOR THE RETAINED MC SAMPLES. SENS. IS SENSITIVITY AND SPEC. IS SPECIFICITY. THESE ARE IMAGING-BURDEN SCORES, NOT PATHOLOGY RESPONSE PROBABILITIES.

Readout	AUC	Sens.	Spec.	PPV/NPV
$P(T_3\,\mathrm{FTV} < 5)$	0.957	0.755	0.970	0.870/0.937
Last FTV < 5 mL	0.961	0.679	0.980	0.900/0.920
$P(T_3\,\mathrm{FTV} > 20)$	0.984	0.967	0.914	0.844/0.983
Last FTV > 20 mL	0.986	0.972	0.902	0.827/0.985

V. DISCUSSION

The results support a layered interpretation of longitudinal tumor forecasting. Graph message passing is valuable when local neighborhoods reveal response structure that isolated node features only partially observe. Endpoint calibration is separately necessary because a graph rollout can preserve sampled-cloud geometry while still needing correction in total active burden. Residual MC then converts the calibrated center into a patient-level distribution whose coverage and sharpness can be evaluated directly.

The main result is therefore not a single model ranking. It is a modular evaluation showing that graph dynamics, endpoint calibration, scalar readout calibration, and residual uncertainty calibration have different roles. This is why the scalar controls, transfer stress test, and tail analysis are part of the evidence rather than secondary caveats.

The endpoint-calibration result is also a methodological

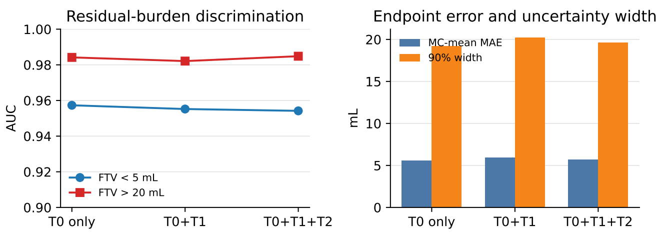

Fig. 8. Serial MRI-burden monitoring readouts from the retained model. Low- and high-burden discrimination remain high as additional MRI visits are used for conditioning, while MC error and interval width summarize endpoint uncertainty.

caution in a useful direction. A future tumor graph can score well in sampled-cloud metrics while still requiring correction in total active burden. Conversely, a scalar FTV center can be accurate while discarding the local tumor state entirely. Treating FTV as a graph readout rather than an independent regression head keeps these two targets connected: the endpoint loss corrects the clinical burden center, while the active-burden term keeps the correction attached to the predicted node activity. This is the reason the retained model is evaluated on active-node error, SWD, Chamfer, Dice, coverage, width, and CRPS rather than on FTV MAE alone.

For breast imaging, the most defensible use is uncertainty-aware MRI-burden monitoring. Expected FTV summarizes likely residual enhancing burden; interval width and coverage describe uncertainty; and MC threshold probabilities give readouts such as low residual burden or substantial residual burden. These outputs could support trial analytics or reader-facing response monitoring after external validation, but they should not be interpreted as treatment counterfactuals or pathology-response probabilities. The modest association of the low-burden readout with pathologic complete response (pCR) in available labels reinforces that boundary.

The scalar controls are deliberately strong and shape the claim. If the only goal is scalar future FTV, last-observed or hybrid residual calibration is hard to beat. This strengthens the modular interpretation: scalar readouts are efficient for scalar endpoint forecasting, while the graph model supplies the structured tumor state, active-node dynamics, and sampled-cloud geometry needed for spatial monitoring.

This boundary is important for translation. A scalar readout may be preferred for a narrow endpoint-screening task because it is simpler, more transparent, and already competitive for FTV. The graph rollout is justified when the forecasted object itself matters: for example, when downstream analysis needs regional active burden, spatial uncertainty, graph-state trajectories, or patient-specific visualization of plausible future tumor support. The hybrid graph-plus-scalar control is useful in this respect because it shows that the graph center and scalar history can be complementary for scalar calibration without eliminating the need to evaluate the graph-native output separately.

The external stress test should be read in the same conservative way. The Breast-MRI-NACT-Pilot cohort is independent, but the graph-ready subset is small and no target-cohort residual recalibration is performed. The result therefore does not establish external clinical generalization. It does show that the retained edge-aware graph remains the best of the tested graph-family variants under a collection shift, while the absolute coverage and bias expose the need for recalibration before deployment. That transfer boundary is useful because it points to the next validation requirement rather than hiding transfer risk behind internal held-out performance.

The reproducibility package is designed around the same constraints. Raw DCE-MRI volumes and protected patient-level imaging materials are not redistributed, but the release records the training and evaluation scripts, retained configurations, aggregate paper tables, figure-generation code, and data-access instructions needed to audit the claims. This is especially important because the main manuscript must remain self-contained: implementation details that affect interpretation, such as held-out-patient residual exclusion, the retained edge design, and comparator roles, are stated in the paper rather than delegated to a manuscript-style supplement.

The main limits are also explicit boundary conditions. The I-SPY2/ACRIN result is an internal held-out-patient evaluation, and the Breast-MRI-NACT-Pilot analysis is a small stress test rather than a powered external validation. The high-volume tail is not hidden by aggregate calibration; it defines the current safety boundary for clinical interpretation. Future work should test larger independent cohorts, FTV-stratified and subtype-aware residual calibration, explicit decomposition of spatial and scalar uncertainty, and prospective studies of whether uncertainty-aware FTV trajectories change reader confidence or trial-level response assessment. The most direct technical extension is not simply a larger graph model, but calibration that recognizes when a patient is entering the high-burden tail or a shifted acquisition regime where source residual pools are no longer reliable.

VI. Conclusion

Endpoint-calibrated spatial graph rollouts convert serial breast DCE-MRI tumor graphs into probabilistic future FTV distributions while retaining a graph-native tumor-state forecast. In held-out I-SPY2/ACRIN patients, the retained Hybrid-Edge $k = 8$ graph reduced deterministic FTV MAE from 12.88 to 4.63 mL, graph-based MC-mean FTV MAE from 15.26 to 5.56 mL, raw 90% coverage from 0.766 to 0.877, and conformal 90% interval width from 59.51 to 22.38 mL. Scalar and hybrid controls remain strongest for scalar-only FTV, but the graph model provides the structured spatial state needed for longitudinal tumor-state forecasting. The central methodological result is that graph dynamics, endpoint calibration, and residual uncertainty calibration are separable layers that should be evaluated together.

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